

Tide retrospective: the stable recovery interface (smooth-germ programme, stage C1)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The smooth-germ programme’s named trap is that replacing a smooth loss by its Taylor polynomial produces *order-controlled* moment errors, never exact equality — so the polynomial recovery theorems, whose public forms take eventually-equal data, cannot be invoked directly. The scoping consult prescribed stage C1: a theorem interface in which every data hypothesis is a vanishing limit at the coefficient-sensitive scale.

2 The thing formalised

Two theorems in `Laplace/OneD/StableRecovery.lean`, sorry-free with zero warnings on the first build¹: variable-base recovery from pure `Tendsto` data — the second-moment difference tends to 0, and each rung’s scaled moment difference $q^{-r}(F_{2k+r}^1 - F_{2k+r}^2)$ tends to 0 — concluding $a_1 = a_2$ and $c^1 = c^2$; and its $k = 1$ form in the note’s $\lambda/2$ normalisation.

3 Why it was nearly free

The consult predicted the existing proofs would “adapt”; in fact no adaptation was needed at all. The programme’s internal theorems were already limit-based — the strong-induction recovery takes scaled `Tendsto` hypotheses, and the base recovery was built with an `_of_tendsto` variant precisely as the bridge — so C1 is eleven lines of composition. Only the outermost public packaging had hardened its data to eventual equality (for the t -transfer, where exact equality is natural). The moral, worth keeping: maintaining the limit-based interface alongside the equality interface at every layer is what makes a stability programme cheap later.

4 Deliberation and check

Prior-deliberation path (C1 is specified verbatim in the archived scoping consult); no distinct numerical check is feasible or needed — the statement is an interface weakening of theorems whose analytic content was checked in the two preceding tides, and it adds no closed form.

¹`jet_recovery_stable` and `nondegenerateJet_recovery_stable`.

5 What the next tide inherits

Stages C2–C6 are the genuinely analytic remainder: two-sided quadratic control near the minimum and domain splitting (C2), rescaled Taylor-remainder bounds on the moving inner region (C3), normalized-quotient and cutoff transfer (C4), the finite-order smooth recovery (C5), and full-jet quantifier packaging (C6). These need a dedicated scoping consult with the note’s “Justification” paragraph as the primary source; C2’s outer estimate in particular must decide what global hypothesis stands in for the note’s quadratic bound.